Bhinu Puvvala

Madison, WI | +1 (312) 918-4861 | bhinusantosh@gmail.com | linkedin.com/in/bhinu-puvvala | github.com/bhinu

Education

Bachelor of Science - University of Wisconsin–Madison | *Madison, WI* *Expected May 2027*

- **Major**: Computer Science, Data Science; **Minor**: Economic Analytics; **GPA**: 3.76/4.0; Dean's List
- **Relevant Coursework**: Data Structures and Algorithms, Operating Systems, Distributed Systems, Database Systems, Computer Vision

Experience

Software Engineer, Industry Capstone - Holos | *Madison, WI* *Sep 2026 – Present*

- Building a **Python** data pipeline, with CI in **Azure DevOps**, that converts **Apple Vision Pro** expert captures into labeled, robot-ready training data for the robotics team
- Developing **action segmentation** and **grasp classification** models that split 52-point bimanual hand captures into sub-actions (reach, grasp, insert, fasten, release) via kinematics and object motion to improve robot task accuracy

Software Engineering Intern - QubitSolve (NSF-funded) | *Morgantown, WV* *Jun 2026 – Aug 2026*

- Built and launched a **Python** cost-estimation engine for QubitFlux's quantum CFD product, pricing runs across **AWS EC2** and **5 quantum backends** via a **FastAPI** app that cut sales quotes from minutes to seconds
- Cut unexpected cloud spend by **20%+** with per-run budget caps modeling compute, head-node, and **EBS** costs, routing jobs to compute- or memory-optimized **EC2** instances at a **32 GB** threshold
- Fixed IBM cost estimates that always returned **$0.00** by tracing a missing shots multiplier and unpopulated gate counts, then computed counts analytically since **Qiskit** transpilation exceeded the **500 ms** budget
- Cut simulation-parameter setup from **2+ hours** in a spreadsheet to seconds with a recommender that finds stable parameter ranges and, for infeasible inputs, reports the minimum grid size needed

Software Engineering Intern - Methix (Microsoft-backed) | *New York, NY* *May 2025 – Aug 2025*

- Reduced **p95 API latency by 30%** for **5,000+ weekly users** by adding **Redis** caching to **Flask** services on **Azure**
- Lowered **PostgreSQL** query latency by **52%** (250 to 120 ms) by profiling hot endpoints and redesigning indexes
- Cut deploy time by **80%** for a **4-engineer** backend team by rebuilding and owning its **Azure DevOps** pipeline
- Enforced plan-based access for **4 user tiers** across **45+ endpoints** with **JWT** claims, RBAC, and rate limiting

Undergraduate Researcher - GLUE Lab, UW–Madison | *Madison, WI* *Jan 2025 – May 2025*

- Cut inference time by **86%** on **10 TB+** of satellite imagery by parallelizing **PyTorch** inference with **Dask**
- Raised cattle-confinement detection accuracy by **20%** by fine-tuning **YOLOv8** and **CSRNet**

Software Engineering Intern - Diyar United Company | *Kuwait City, Kuwait* *Jun 2024 – Aug 2024*

- Automated **4 Python/SQL ETL pipelines** processing **500K+ records per day**, tripling throughput and saving **15+ hours** of manual work each week
- Designed the **star-schema** warehouse behind those pipelines and built **Power BI** dashboards used by **10K+ clients**

Projects

MuscleMemory: Browser Agent That Learns From Fixes | ***Python, Playwright, FastAPI, SQLite, OpenAI***

- Cut manual interventions by **40%** over **30** simulated days of site changes with an LLM agent that replays human fixes
- Blocked retries of irreversible actions (zero duplicate orders, 10 runs) and caught stale fixes via page landmarks
- Validated LLM-generated fixes against a **pytest** regression suite before replay, rejecting unsafe or broken actions

Videre: On-Device AI Video Editor | ***Snapdragon NPU, ONNX Runtime QNN, Whisper, FFmpeg***

- Won the **Qualcomm track at MadData 2026** (**30 teams**) with an offline, on-device AI video editor
- Ran **Whisper** and **SigLIP2** on the NPU via **ONNX Runtime QNN** for real-time, word-level transcription

Technical Skills

Languages: Python, C/C++, JavaScript, SQL, Bash
Backend & Cloud: FastAPI, Flask, REST APIs, PostgreSQL, Redis (NoSQL), Docker, AWS (EC2, EBS), Azure
ML & Data: PyTorch, ONNX Runtime, Qiskit, YOLOv8, Dask, GeoPandas, Pandas, NumPy, Power BI
Tools: Git, CI/CD, Azure DevOps, pytest, Linux, React, Playwright, Cursor, Claude Code, Codex